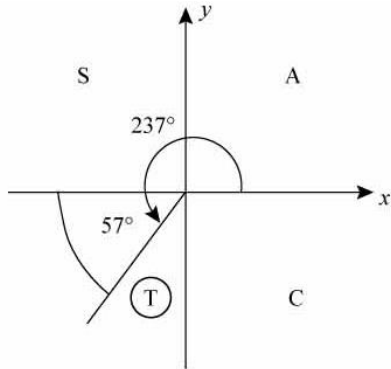
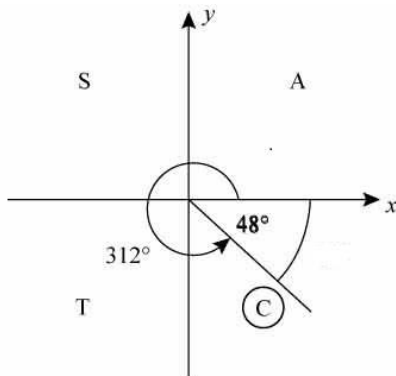


## Trigonometric identities and equations, Mixed exercise 10

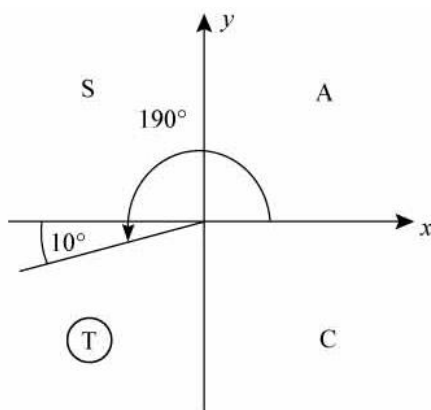
- 1 a  $237^\circ$  is in the third quadrant, so  $\cos 237^\circ$  is -ve.  
The angle made with the horizontal is  $57^\circ$ .  
So  $\cos 237^\circ = -\cos 57^\circ$



- b  $312^\circ$  is in the fourth quadrant so  $\sin 312^\circ$  is -ve.  
The angle to the horizontal is  $48^\circ$ .  
So  $\sin 312^\circ = -\sin 48^\circ$



- c  $190^\circ$  is in the third quadrant so  $\tan 190^\circ$  is +ve.  
The angle to the horizontal is  $10^\circ$ .  
So  $\tan 190^\circ = +\tan 10^\circ$



2 a  $\cos 270^\circ = 0$

b  $\sin 225^\circ = \sin(180 + 45)^\circ$   
 $= -\sin 45^\circ$   
 $= -\frac{\sqrt{2}}{2}$

c  $\cos 180^\circ = -1$  (see graph of  $y = \cos \theta$ )

d  $\tan 240^\circ = \tan(180 + 60)^\circ$   
 $= +\tan 60^\circ$  (third quadrant)  
So  $\tan 240^\circ = +\sqrt{3}$

e  $\tan 135^\circ = -\tan 45^\circ$  (second quadrant)  
So  $\tan 135^\circ = -1$

3 Using  $\sin^2 A + \cos^2 A \equiv 1$

$$\sin^2 A + \left(-\sqrt{\frac{7}{11}}\right)^2 = 1$$

$$\sin^2 A = 1 - \frac{7}{11}$$

$$= \frac{4}{11}$$

$$\sin A = \pm \frac{2}{\sqrt{11}}$$

But  $A$  is in the second quadrant (obtuse),

so  $\sin A$  is +ve.

$$\text{So } \sin A = +\frac{2}{\sqrt{11}}$$

$$\text{Using } \tan A = \frac{\sin A}{\cos A}$$

$$\tan A = \frac{\left(\frac{2}{\sqrt{11}}\right)}{-\sqrt{\frac{7}{11}}}$$

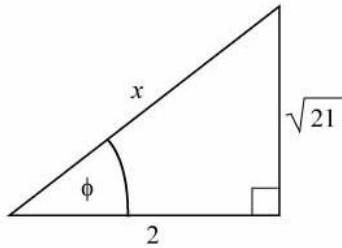
$$= -\frac{2}{\sqrt{11}} \times \frac{\sqrt{11}}{\sqrt{7}}$$

$$= -\frac{2}{\sqrt{7}}$$

$$= -\frac{2\sqrt{7}}{7}$$

(rationalising the denominator)

- 4 Draw a right-angled triangle with an angle of  $\phi$ , where  $\phi = +\frac{\sqrt{21}}{2}$ .



Use Pythagoras' theorem to find the hypotenuse.

$$\begin{aligned}x^2 &= 2^2 + (\sqrt{21})^2 \\&= 4 + 21 \\&= 25\end{aligned}$$

So  $x = 5$

a  $\sin \phi = \frac{\sqrt{21}}{5}$

As  $B$  is reflex and  $\tan B$  is +ve,  $B$  is in the third quadrant.

So  $\sin B = -\sin \phi$

$$= -\frac{\sqrt{21}}{5}$$

b From the diagram  $\cos \phi = \frac{2}{5}$ .

$B$  is in the third quadrant. So  $\cos B = -\cos \phi$

$$= -\frac{2}{5}$$

- 5 a Factorise  $\cos^4 \theta - \sin^4 \theta$ .  
(This is the difference of two squares.)

$$\begin{aligned}\cos^4 \theta - \sin^4 \theta &= (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta) \\&= (1)(\cos^2 \theta - \sin^2 \theta) \\&\quad (\text{as } \cos^2 \theta + \sin^2 \theta \equiv 1)\end{aligned}$$

$$\text{So } \cos^4 \theta - \sin^4 \theta = \cos^2 \theta - \sin^2 \theta$$

- b Factorise  $\sin^2 3\theta - \sin^2 3\theta \cos^2 3\theta$ .
- $$\begin{aligned}\sin^2 3\theta - \sin^2 3\theta \cos^2 3\theta \\&= \sin^2 3\theta (1 - \cos^2 3\theta)\end{aligned}$$

- 5 b (use  $\sin^2 3\theta + \cos^2 3\theta \equiv 1$ )

$$\begin{aligned}\sin^2 3\theta - \sin^2 3\theta \cos^2 3\theta &= \sin^2 3\theta (\sin^2 3\theta) \\&= \sin^4 3\theta\end{aligned}$$

c  $\cos^4 \theta + 2\sin^2 \theta \cos^2 \theta + \sin^4 \theta$

$$\begin{aligned}&= (\cos^2 \theta + \sin^2 \theta)^2 \\&= 1 \quad (\text{since } \sin^2 \theta + \cos^2 \theta \equiv 1)\end{aligned}$$

- 6 a  $2(\sin x + 2\cos x) = \sin x + 5\cos x$
- $$\begin{aligned}\Rightarrow 2\sin x + 4\cos x &= \sin x + 5\cos x \\ \Rightarrow 2\sin x - \sin x &= 5\cos x - 4\cos x \\ \Rightarrow \sin x &= \cos x\end{aligned}$$

(divide both sides by  $\cos x$ )

$$\text{So } \tan x = 1$$

b  $\sin x \cos y + 3\cos x \sin y$

$$\begin{aligned}&= 2\sin x \sin y - 4\cos x \cos y \\ \Rightarrow \frac{\sin x \cos y}{\cos x \cos y} + \frac{3\cos x \sin y}{\cos x \cos y} \\&= \frac{2\sin x \sin y}{\cos x \cos y} - \frac{4\cos x \cos y}{\cos x \cos y} \\ \Rightarrow \tan x + 3\tan y &= 2\tan x \tan y - 4 \\ \Rightarrow 2\tan x \tan y - 3\tan y &= 4 + \tan x \\ \Rightarrow \tan y(2\tan x - 3) &= 4 + \tan x \\ \text{So } \tan y &= \frac{4 + \tan x}{2\tan x - 3}\end{aligned}$$

- 7 a LHS =  $(1 + 2\sin \theta + \sin^2 \theta) + \cos^2 \theta$
- $$\begin{aligned}&= 1 + 2\sin \theta + 1 \quad \text{since } \sin^2 \theta + \cos^2 \theta \equiv 1 \\&= 2 + 2\sin \theta \\&= 2(1 + \sin \theta) \\&= \text{RHS}\end{aligned}$$

b LHS =  $\cos^4 \theta + \sin^2 \theta$

$$\begin{aligned}
 &= (\cos^2 \theta)^2 + \sin^2 \theta \\
 &= (1 - \sin^2 \theta)^2 + \sin^2 \theta \\
 &= 1 - 2\sin^2 \theta + \sin^4 \theta + \sin^2 \theta \\
 &= (1 - \sin^2 \theta) + \sin^4 \theta \\
 &= \cos^2 \theta + \sin^4 \theta \\
 &\quad (\text{using } \sin^2 \theta + \cos^2 \theta \equiv 1) \\
 &= \text{RHS}
 \end{aligned}$$

**8 a**  $\sin \theta = \frac{3}{2}$  has no solutions as  
 $-1 \leq \sin \theta \leq 1$ .

**b**  $\sin \theta = -\cos \theta$   
 $\Rightarrow \tan \theta = -1$

Look at the graph of  $y = \tan \theta$  in the interval  $0 \leq \theta \leq 360^\circ$ . There are two solutions.

- c** The minimum value of  $2 \sin \theta$  is  $-2$ .  
 The minimum value of  $3 \cos \theta$  is  $-3$ .  
 Each minimum value is for a different  $\theta$ .  
 So the minimum value of  
 $2 \sin \theta + 3 \cos \theta$  is always greater than  $-5$ .  
 There are no solutions of  
 $2 \sin \theta + 3 \cos \theta + 6 = 0$   
 as the LHS can never be zero.

- d** Solving  $\tan \theta + \frac{1}{\tan \theta} = 0$  is equivalent to  
 solving  $\tan^2 \theta = -1$ , which has no  
 solutions.  
 So there are no solutions.

**9 a**  $4xy - y^2 + 4x - y \equiv y(4x - y) + (4x - y)$   
 $= (4x - y)(y + 1)$

**b** Using **a** with  $x = \sin \theta$ ,  $y = \cos \theta$   
 $4 \sin \theta \cos \theta - \cos^2 \theta$   
 $+ 4 \sin \theta - \cos \theta = 0$   
 So  
 $(4 \sin \theta - \cos \theta)(\cos \theta + 1) = 0$

So  $4 \sin \theta - \cos \theta = 0$  or

$\cos \theta + 1 = 0$

$4 \sin \theta - \cos \theta = 0$

$\Rightarrow \tan \theta = \frac{1}{4}$

The calculator solution is  $\theta = 14.0^\circ$ .

$\tan \theta$  is +ve so  $\theta$  is in the first and third quadrants.

So  $\theta = 14.0^\circ, 194^\circ$

$\cos \theta + 1 = 0 \Rightarrow \cos \theta = -1$

So  $\theta = +180^\circ$  (from graph)

Solutions are  $\theta = 14.0^\circ, 180^\circ, 194^\circ$

**10 a** As  $\sin(90 - \theta)^\circ \equiv \cos \theta^\circ$ ,  
 $\sin(90 - 3\theta)^\circ \equiv \cos 3\theta^\circ$   
 So  $4\cos 3\theta^\circ - \sin(90 - 3\theta)^\circ$   
 $= 4\cos 3\theta^\circ - \cos 3\theta^\circ$   
 $= 3\cos 3\theta^\circ$

**b** Using **a**,  $4\cos 3\theta^\circ - \sin(90 - 3\theta)^\circ = 2$   
 is equivalent to  $3\cos 3\theta^\circ = 2$

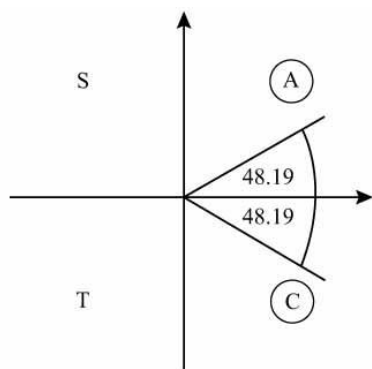
so  $\cos 3\theta^\circ = \frac{2}{3}$

Let  $X = 3\theta$  and solve  $\cos X^\circ = \frac{2}{3}$

in the interval  $0^\circ \leq X \leq 1080^\circ$ .

The calculator solution is  $X = 48.19^\circ$

As  $\cos X^\circ$  is +ve,  $X$  is in the first and fourth quadrants.



Read off all solutions in the interval  $0^\circ \leq X \leq 1080^\circ$ .

$X = 48.19^\circ, 311.81^\circ, 408.19^\circ, 671.81^\circ, 768.19^\circ, 1031.81^\circ$

So  $\theta = \frac{X}{3} = 16.1^\circ, 104, 136^\circ, 224^\circ, 256^\circ, 344^\circ$  (3 s.f.)

**11 a**  $2\sin 2\theta = \cos 2\theta$

$\Rightarrow \frac{2\sin 2\theta}{\cos 2\theta} = 1$

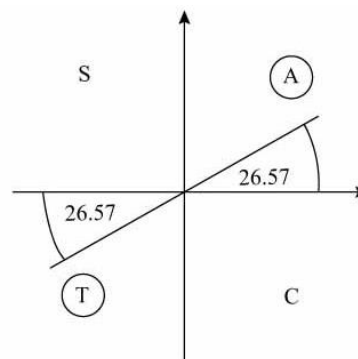
$\Rightarrow 2\tan 2\theta = 1$  since  $\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta}$

So  $\tan 2\theta = 0.5$

**b** Solve  $\tan 2\theta^\circ = 0.5$  in the interval  $0^\circ \leq \theta < 360^\circ$  or  $\tan X^\circ = 0.5$  where  $X = 2\theta$ ,  $0^\circ \leq X < 720^\circ$ .

**11 b** The calculator solution for  $\tan^{-1} 0.5$  is  $26.57^\circ$ .

As  $\tan X$  is +ve,  $X$  is in the first and third quadrants.



Read off solutions for  $X$  in the interval  $0^\circ \leq X < 720^\circ$ .

$X = 26.57^\circ, 206.57^\circ, 386.57^\circ, 566.57^\circ$   
 $= 2\theta$

So  $\theta = \frac{X}{2}$

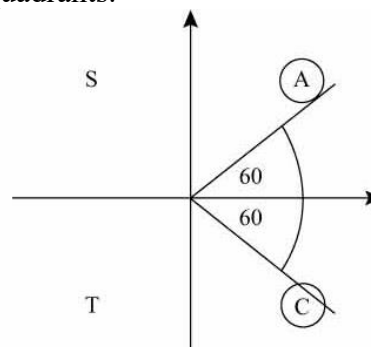
$= 13.3^\circ, 103.3^\circ, 193.3^\circ, 283.3^\circ$  (1 d.p.)

**12 a**  $\cos(\theta + 75)^\circ = 0.5$

Solve  $\cos X^\circ = 0.5$ , where  $X = \theta + 75$ ,  
 $75^\circ \leq X < 435^\circ$ .

Your calculator solution for  $X = 60^\circ$ .

As  $\cos X$  is +ve,  $X$  is in the first and fourth quadrants.



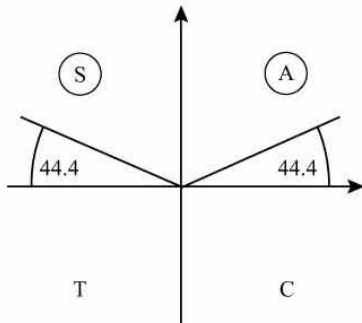
Read off all solutions in the interval  $75^\circ \leq X < 435^\circ$ .

$X = 300^\circ, 420^\circ$

$\theta + 75^\circ = 300^\circ, 420^\circ$

So  $\theta = 225^\circ, 345^\circ$

- 12 b**  $\sin 2\theta = 0.7$  in the interval  $0^\circ \leq \theta < 360^\circ$ .  
Solve  $\sin X^\circ = 0.7$ , where  
 $X = 2\theta$ ,  $0^\circ \leq X < 720^\circ$ .  
The calculator solution is  $44.4^\circ$ .  
As  $\sin X$  is +ve,  $X$  is in the first and second quadrants.



Read off solutions in the interval  $0^\circ \leq X < 720^\circ$ .  
 $X = 44.4^\circ, 135.6^\circ, 404.4^\circ, 495.6^\circ$   
 $= 2\theta$   
So  $\theta = \frac{X}{2}$   
 $= 22.2^\circ, 67.8^\circ, 202.2^\circ, 247.8^\circ$  (1 d.p.)

- 13** Multiply both sides of the equation by  $(1 - \cos 2x)$ , provided  $\cos 2x \neq 1$ .  
*Note:* In the interval given,  $\cos 2x$  is never equal to 1.  
So  $\cos 2x + 0.5 = 2 - 2 \cos 2x$

$$\Rightarrow 3 \cos 2x = \frac{3}{2}$$

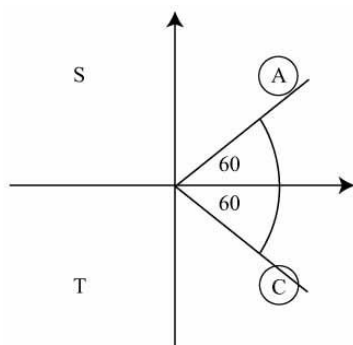
$$\text{So } \cos 2x = \frac{1}{2}$$

$$\text{Solve } \cos X = \frac{1}{2} \text{ where } X = 2x,$$

$$0^\circ < X < 540^\circ.$$

The calculator solution is  $60^\circ$ .

As  $\cos X$  is +ve,  $X$  is in the first and fourth quadrants.



- 13** Read off solutions for  $X$  in the interval  $0^\circ < X < 540^\circ$ .  
 $X = 60^\circ, 300^\circ, 420^\circ$

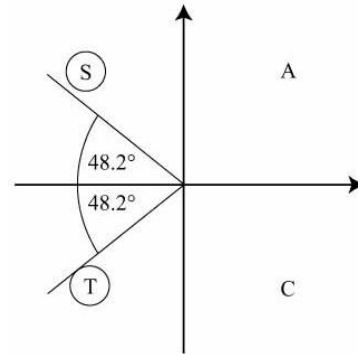
$$\text{So } x = \frac{X}{2}$$

$$= 30^\circ, 150^\circ, 210^\circ$$

- 14** Using  $\sin^2 \theta + \cos^2 \theta \equiv 1$   
 $2 \cos^2 \theta - \cos \theta - 1 = 1 - \cos^2 \theta$   
 $\Rightarrow 3 \cos^2 \theta - \cos \theta - 2 = 0$   
 $\Rightarrow (3 \cos \theta + 2)(\cos \theta - 1) = 0$   
So  $3 \cos \theta + 2 = 0$  or  $\cos \theta - 1 = 0$   
For  $3 \cos \theta + 2 = 0$ ,  $\cos \theta = -\frac{2}{3}$

The calculator solution is  $131.8^\circ$ .

As  $\cos \theta$  is -ve,  $\theta$  is in the second and third quadrants.



So solutions are

$$\theta = 131.8^\circ, 228.2^\circ$$

- 15 a** The student found additional solutions after dividing by three rather than before. The student has not applied the full interval for the solutions.

**b** Let  $X = 3x$

$$\sin X = \frac{1}{2}$$

As  $X = 3x$ , then as  $-360^\circ \leq x \leq 360^\circ$

$$\text{So } 3 \times -360^\circ \leq X \leq 3 \times 360^\circ$$

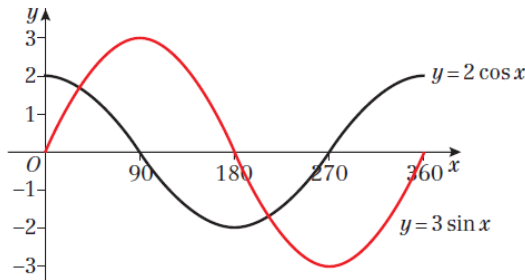
So the interval for  $X$  is

$$-1080^\circ \leq X \leq 1080^\circ$$

- 15 b**  $X = 30^\circ, 150^\circ, 390^\circ, 510^\circ, 750^\circ, 870^\circ, -210^\circ, -330^\circ, -570^\circ, -690^\circ, -930^\circ, -1050^\circ$

i.e.  $3x = -1050^\circ, -930^\circ, -690^\circ, -570^\circ,$   
 $-330^\circ, -210^\circ, 30^\circ, 150^\circ, 390^\circ, 510^\circ,$   
 $750^\circ, 870^\circ$   
 So  $x = -350^\circ, -310^\circ, -230^\circ, -190^\circ,$   
 $-110^\circ, -70^\circ, 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ,$   
 $290^\circ$

16 a



b The graphs intersect at two places so there are two solutions to the equation in the given range.

c  $3\sin x = 2\cos x$   
 $\frac{\sin x}{\cos x} = \frac{2}{3}$   
 $\tan x = \frac{2}{3}$   
 $x = 33.7^\circ, 213.7^\circ$

17 a Using the cosine rule

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

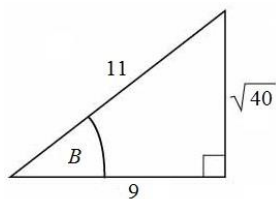
$$\cos B = \frac{6^2 + 11^2 - 7^2}{2 \times 6 \times 11}$$

$$\cos B = \frac{36 + 121 - 49}{132}$$

$$\cos B = \frac{9}{11}$$

b Using Pythagoras' theorem

$$\sqrt{11^2 - 9^2} = \sqrt{40}$$



$$\sin B = \frac{\sqrt{40}}{11} = \frac{2\sqrt{10}}{11}$$

18 a Using the sine rule

$$\frac{\sin Q}{q} = \frac{\sin P}{p}$$

$$\frac{\sin Q}{6} = \frac{\sin 45^\circ}{5}$$

$$\sin Q = \frac{6\left(\frac{\sqrt{2}}{2}\right)}{5}$$

$$\sin Q = \frac{3\sqrt{2}}{5}$$

b Using Pythagoras' theorem or identity  
 $\cos^2 x + \sin^2 x = 1$

$$\cos Q = \frac{\sqrt{7}}{5} \text{ for the acute angle}$$

As  $Q$  is obtuse, it is in the second quadrant where  $\cos Q$  is negative.

$$\text{So } \cos Q = -\frac{\sqrt{7}}{5}$$

19 a  $3\sin^2 x - \cos^2 x = 2$  can be written as  
 $3\sin^2 x - (1 - \sin^2 x) = 2$   
 which reduces to  
 $4\sin^2 x = 3$

b  $4\sin^2 x = 3$

$$\sin^2 x = \frac{3}{4}$$

$$\sin x = \pm \frac{\sqrt{3}}{2}$$

$$x = 60^\circ, 120^\circ, -60^\circ, -120^\circ$$

So the solutions are

$$x = -120.0^\circ, -60.0^\circ, 60.0^\circ, 120.0^\circ$$

20  $3\cos^2 x + 1 = 4\sin x$  can be written as

$$3(1 - \sin^2 x) + 1 = 4\sin x$$

which can be reduced to

$$3\sin^2 x + 4\sin x - 4 = 0$$

$$(3\sin x - 2)(\sin x + 2) = 0$$

$$\sin x = \frac{2}{3} \text{ or } \sin x = -2$$

$\sin x = -2$  has no solutions.

$$\text{So } x = 41.8^\circ, 138.2^\circ, -221.8^\circ, -318.2^\circ$$

So the solutions are

$$x = -318.2^\circ, -221.8^\circ, 41.8^\circ, 138.2^\circ$$

**21 a** Solving equation where  $X = 2x + k$

$$3 + \sqrt{3} = 3 + 2 \sin(X)$$

$$\sqrt{3} = 2 \sin(X)$$

$$X = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$X = 60^\circ \text{ or } 180^\circ - 60^\circ = 120^\circ$$

$$k = 60^\circ - 30^\circ \text{ or } 120^\circ - 30^\circ$$

$$k = 30^\circ \text{ or } 90^\circ$$

**b** Solving equation where  $X = 2x + 30$

and  $30 \leq X \leq 750$

$$1 = 3 + 2 \sin(X)$$

$$-2 = 2 \sin(X)$$

$$X = \sin^{-1}(-1)$$

$$X = 270^\circ \text{ or } 270^\circ + 360^\circ = 630^\circ$$

$$x = \frac{X - 30^\circ}{2}$$

$$x = 120^\circ \text{ or } 300^\circ$$

### Challenge

$$\tan^4 x - 3 \tan^2 x + 2 = 0$$

$$(\tan^2 x - 2)(\tan^2 x - 1) = 0$$

$$\tan^2 x = 2 \text{ or } \tan^2 x = 1$$

$$\tan x = \pm 1 \text{ or } \pm \sqrt{2}$$

$$x = 45^\circ, 225^\circ, -45^\circ, 135^\circ, 315^\circ, 54.7^\circ, 234.7^\circ, -54.7^\circ, 125.3^\circ, 305.3^\circ$$

So the solutions are

$$x = 45^\circ, 54.7^\circ, 125.3^\circ, 135^\circ, 225^\circ, 234.7^\circ, 305.3^\circ, 315^\circ$$